WEEKLY LESSON PLAN · 2026-27

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 · Advanced · Senior · Mr. Gavaldon · Da Vinci School
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SEMESTER 1
Week 2

MONDAY — Day 6 - Launch: Engineering Firms + Pick Your Problem

Aug 17

OBJECTIVE
Students will form engineering firms, choose a real problem, and define why it matters (Design Brief p.1).

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - industry-standard CAD (this is the shared Semester-1 Fusion foundation)

ELPS / EB SUPPORT
ELPS/EB supports at-level: every constraint symbol shown on-screen and on a station card (VC, DP); pre-teach parametric / constraint / degree-of-freedom with a picture wall before the build (PV); directions restated in short numbered steps (CD, RS); fully-defined finishers coach a partner in Spanish or English (PN); wait time after the DOF question (WT).

AGENDA · 45 MIN
- 6′ Bell-work (format + example): name a badly-designed everyday thing
- 10′ Launch: the firm mission + week map
- 8′ Worked example: a firm end-to-end
- 6′ Randomize firms + assign 4 roles
- 15′ Pick your problem + start Design Brief page 1

SLIDES / ACTIVITY
Engineering Firms – Launch

NOTEBOOK
Sketch the Fusion screen and label the toolbar, sketch button, and timeline.

TURN IN
Engineering Firms - Design Brief packet, page 1 (firm, roles, problem, why)

TUESDAY — Day 7 - Define the Problem

Aug 18

OBJECTIVE
Students will define their problem with a stakeholder, evidence, testable requirements, and a success statement (Design Brief p.2).

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - CAD - solid modeling (shared Fusion foundation)

ELPS / EB SUPPORT
ELPS/EB supports at-level: live demo of each feature before work time with the icon shown (VC, RS); one-page step card per station (CD); pre-teach extrude / fillet / chamfer / density with a labeled diagram (PV, DP); the mass calculation modeled once, numbers left on the board (RS); partners check each other's arithmetic (PN); wait time on the exit question (WT).

AGENDA · 45 MIN
- 6′ Bell-work: who has it + how you know
- 12′ Define like an engineer: who, evidence, requirements, success
- 10′ Worked example + vocab
- 17′ Write 3-5 testable requirements - Design Brief page 2

SLIDES / ACTIVITY
Engineering Firms – Define It

NOTEBOOK
List the 3 steps you used to turn a sketch into a solid.

TURN IN
Engineering Firms - Design Brief packet, page 2 (definition + requirements)

WEDNESDAY

Day 8 - Design the Solution + AI Render

Aug 19

OBJECTIVE

Students will sketch multiple solutions, choose one against their requirements, and generate an AI product image (Design Brief p.3).

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - CAD - model a real object to dimension (shared Fusion foundation)

ELPS / EB SUPPORT

ELPS/EB supports at-level: the decision matrix modeled column-by-column on the board (VC, RS); pre-teach criteria / weight / percent error / verification (PV, DP); caliper technique demonstrated then practiced with wait time (WT); step card for the measure-model-compare loop (CD); partners measure and cross-check three times (PN).

AGENDA · 45 MIN

- 6′ Bell-work: why sketch more than one idea
- 10′ Sketch 3 solutions + choose with a reason/trade-off
- 10′ Write the product-image prompt + generate
- 19′ Finish Design Brief page 3 (sketches, choice, image)

SLIDES / ACTIVITY

Engineering Firms – Design It

NOTEBOOK

Sketch your object with real dimensions before you model it.

TURN IN

Engineering Firms - Design Brief packet, page 3 (design + AI image)

THURSDAY

Day 9 - Build & Drill the Pitch

Aug 20

OBJECTIVE

Students will build a 5-beat pitch deck with AI and prepare to defend it under questioning (Design Brief p.4).

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - technical communication - presenting a design & its rationale

ELPS / EB SUPPORT

ELPS/EB supports at-level: sentence frames that still demand a number ('I chose ___ because the data showed ___') (CD, RS); pre-teach rationale / trade-off / specification (PV); model brief shown and annotated (VC, DP); rehearse in pairs before whole-class with wait time (PN, WT).

AGENDA · 45 MIN

- 6′ Bell-work: what makes a pitch believable
- 10′ The 5 beats + AI-build the deck from your packet
- 12′ Drill the defense: assign who owns each question
- 17′ Finish deck + one full practice run - Design Brief page 4

SLIDES / ACTIVITY

Engineering Firms – Build the Pitch

NOTEBOOK

Write your 1-minute pitch outline: what it is, why these choices, one thing you'd improve.

TURN IN

Engineering Firms - Design Brief packet, page 4 (pitch deck + Q&A prep)

FRIDAY

Day 10 - Design Review (Capstone)

Aug 21

OBJECTIVE
Student firms will pitch their product and defend it under Q&A from a review panel, scored on a shared rubric.

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - deliver a technical presentation; give & receive feedback

ELPS / EB SUPPORT
ELPS/EB supports at-level: rubric levels shown with a worked example score so evaluation isn't language-gated (VC, DP, CD); feedback frames that force a criterion + number ('Your ___ scored high because ___') (RS); presenters use their rehearsed script (WT); peers may confer before scoring (PN).

AGENDA · 45 MIN
- 6′ Bell-work: what a great audience does
- 6′ Format + rubric (how you are scored)
- 33′ Design Review: 3-min pitch + 2-min defense per firm
- 5′ Exit ticket + Week 3 (30 Days of Fusion) tease

SLIDES / ACTIVITY
Engineering Firms – Design Review

NOTEBOOK
Log your pitch (what went well, one thing to improve) + one piece of feedback you got.

TURN IN
Engineering Firms - Design Review rubric (25 pts) + reflection